CIS 635 Knowledge Discovery & Data Mining

HP Optimization: k-NN, Decision Tree

Decision Tree

nb of legs	weight (lb)	animal
4	2.1	Bunny
4	7	Cat
4	1.7	Bunny
4	9	Cat
4	2.75	Bunny
2	2.5	Chicken
2	3	Chicken

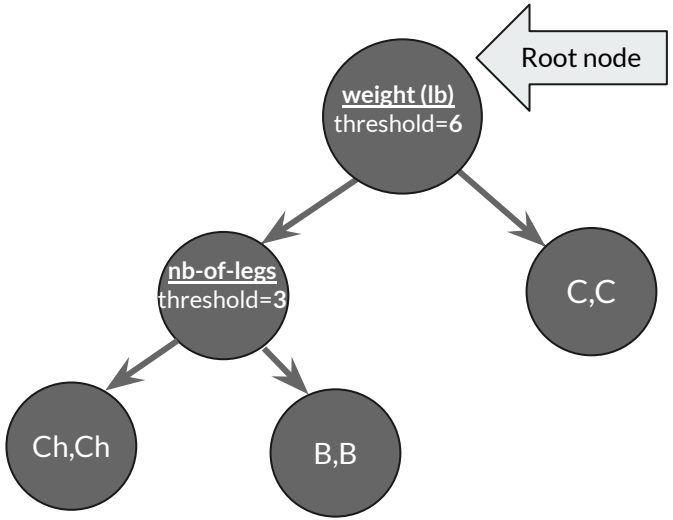
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Test case

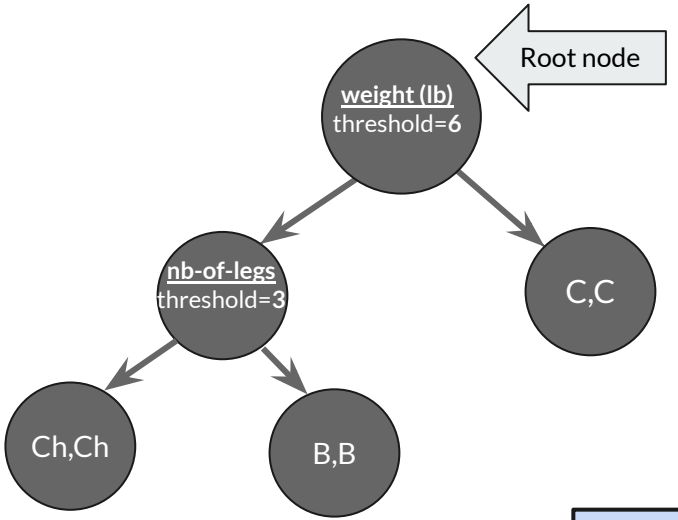
2	4	??

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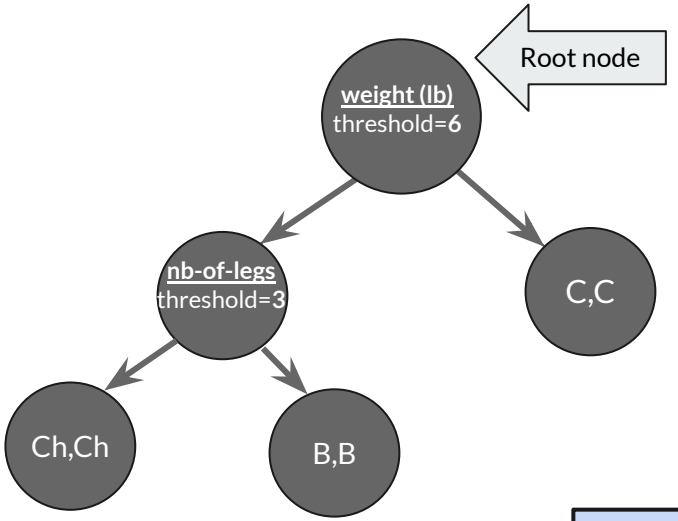
Decision Tree



Test case

2	4	??

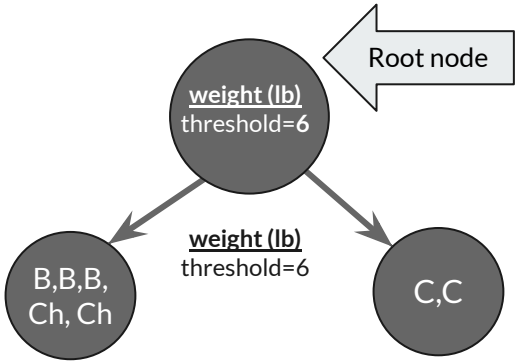
Decision Tree



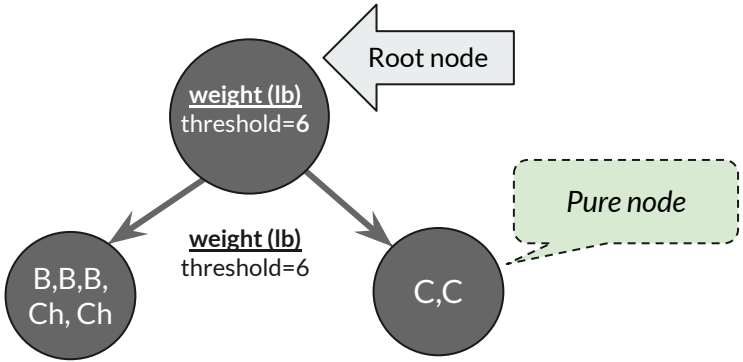
Test case

2	4	Ch

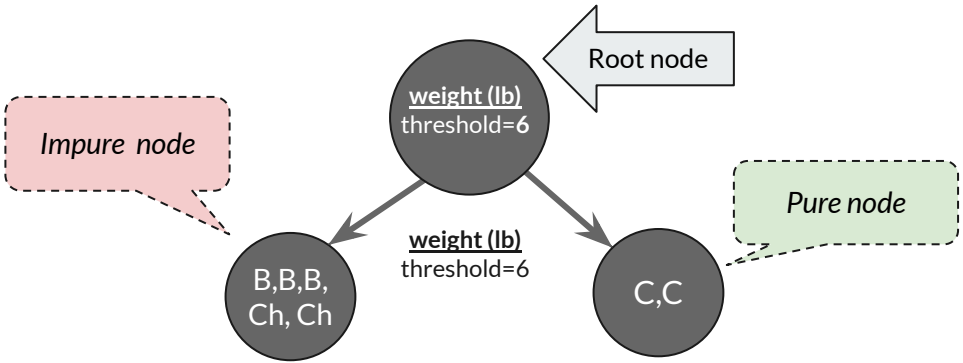
Decision Tree



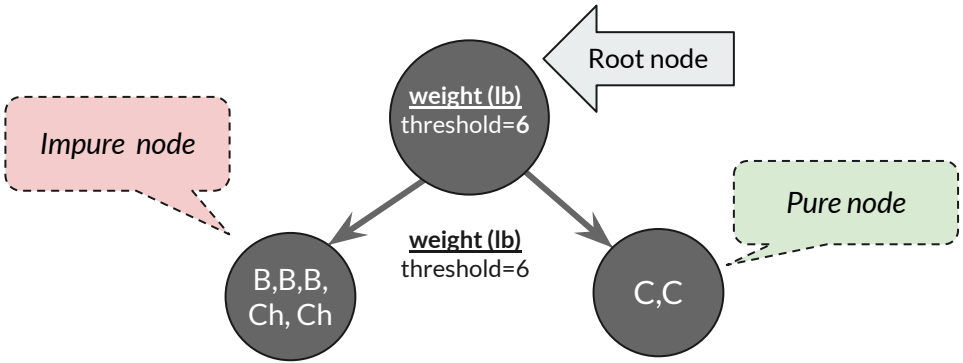
Decision Tree



Decision Tree



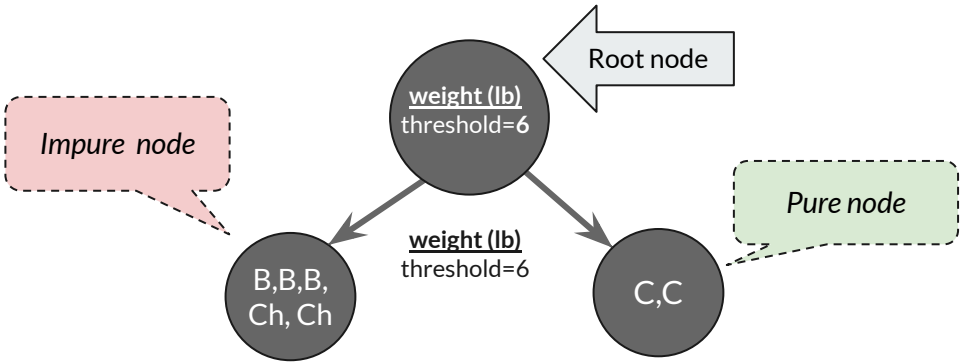
Decision Tree



Test case

2	4	??

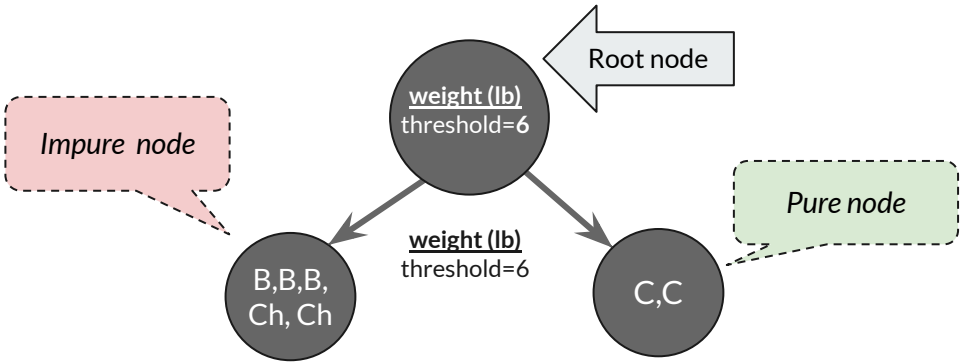
Decision Tree



Test case

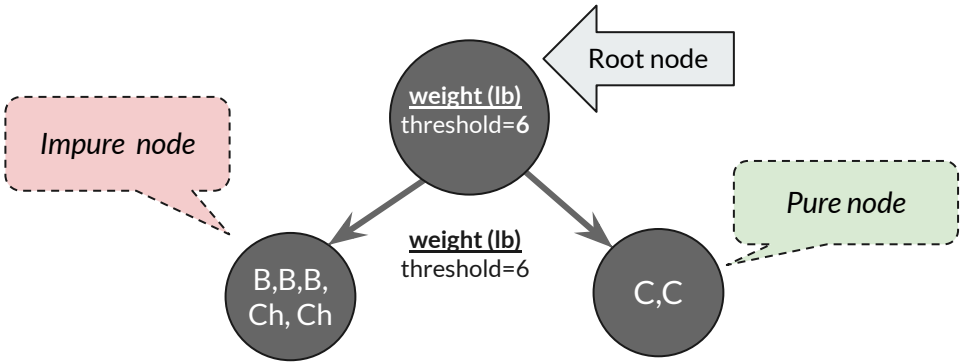
2	??	B

Decision Tree



Never can detect a ***Ch****(icken)*

Decision Tree



Never can detect a ***Ch****(icken)*

But has certain advantages (primarily in the context of ensemble learning, Bagging for an example).

Decision Tree -- Key Hyperparameters

Hyperparameter	Description
criterion	Measure of split quality (e.g., Gini, entropy, squared error).
max_depth	Maximum depth of the tree (controls overfitting).
min_samples_split	Minimum samples required to split an internal node.
min_samples_leaf	Minimum samples required at a leaf node.
max_features	Number of features considered when searching for best split.
max_leaf_nodes	Maximum number of terminal (leaf) nodes.
ccp_alpha	Cost-complexity pruning parameter (larger $\rightarrow$ more pruning).
splitter	Strategy used to choose splits (e.g., best, random).

Random Forest -- Key Hyperparameters

Hyperparameter	Description
n_estimators	Number of trees in the forest.
criterion	Split quality measure used in each tree.
max_depth	Maximum depth of each tree.
min_samples_split	Minimum samples required to split a node.
min_samples_leaf	Minimum samples required at a leaf node.
max_features	Number of features randomly selected per split.
bootstrap	Whether bootstrap sampling is used.
max_samples	Number of samples drawn per tree (if bootstrap=True).
n_jobs	Number of CPU cores used for parallel training.
oob_score	Use out-of-bag samples for generalization estimate.